INDIGENE

STUDY CONSOLE

CSC 101

ULTIMATE MASTER EXAM NOTES

Generations of Computers • Computer Networks & Security • Pseudocode • OSI Model

Memory Hacks + Exam Traps + Rapid-Fire Edition · Large Print

Part	Topic	Core idea
A	Five Generations of Computers	VTIMA: Vacuum, Transistor, IC, Microprocessor, AI
B	Computer Networks	Types, topologies, internetworks, hardware
C	Network Security & Malware	CIA + AAN, virus family, firewall, antivirus
D	Pseudocode	Plain-language algorithm planning
E	OSI Model	Seven layers: Please Do Not Throw Sausage Pizza Away
F	Master Rapid-Fire Sheet	Likely questions with quick answers

Federal University Oye-Ekiti (FUOYE) | 2025/2026 Session

Compiled for: INDIGENE

A THE FIVE GENERATIONS OF COMPUTERS

Each generation = one new core technology → smaller, faster, cheaper

Overview — Why Generations Exist

- The computer evolved from a large, simple calculating machine to a smaller, much more powerful machine.
- Its evolution is defined in terms of **GENERATIONS**.
- Each generation is defined by a new technological development, giving better, cheaper, smaller, faster and more efficient computers.
- There are currently **FIVE** generations, classified by: technology used (hardware/software), computing speed, physical appearance, and applications.

MEMORY HACK: The 5 generations follow ONE repeating pattern: new core technology → smaller size → faster speed → cheaper cost. Remember the core TECH of each generation and everything else follows.

TRAP: Students mix up which generation introduced WHICH technology. The exam loves "which generation introduced X?" — memorize the tech-to-generation pairing as a strict 1-to-1 match.

Quick-Match Table — Tech to Generation (MEMORIZE FIRST)

Generation	Core Technology	Years
1st	Vacuum Tubes	1940–1956
2nd	Transistors	1956–1963
3rd	Integrated Circuits (ICs)	1964–1971
4th	Microprocessor	1971–present
5th	Artificial Intelligence	Present & beyond

> **MEMORY HACK:** VTIMA = Vacuum, Transistor, IC, Microprocessor, AI. Say it like a word: "V-TIM-A". The single most-tested list in this topic.

1ST GENERATION (1940–1956) — Vacuum Tubes

Core Facts

- Used **VACUUM TUBES** for circuitry and **MAGNETIC DRUMS** for memory.
- Often ENORMOUS — taking up entire rooms.
- Relied on **MACHINE LANGUAGE** (lowest-level, hardest language — pure binary).
- Very expensive to operate; used a great deal of electricity and generated a lot of heat, which often caused malfunctions.
- Examples: **UNIVAC** and **ENIAC**.

Advantages

- It was the only electronic device of its kind at the time.
- First device to hold memory.

Disadvantages

- Too bulky (large in size).
- Vacuum tubes burned out frequently.
- Produced a lot of heat.
- Constant maintenance problems.

> **MEMORY HACK:** VUM — Vacuum tubes, Uses a lot of power, Machine language. First gen = the HOTTEST, BIGGEST, HARDEST to program. Picture a room-sized oven that only understands 1s and 0s.

> **TRAP:** Don't confuse UNIVAC/ENIAC (1st gen) with later "small computer" examples — the exam sometimes lists a later-generation device as a distractor. If it fits in a room and needs vacuum tubes, it's 1st gen, full stop.

2ND GENERATION (1956–1963) — Transistors

Core Facts

- **TRANSISTORS** replaced vacuum tubes — the single defining change.
- Moved from cryptic BINARY machine language to **SYMBOLIC** language.
- High-level languages emerged: early **COBOL** and **FORTRAN**.
- The FIRST computers to store their instructions in their own memory.

Advantages

- Size reduced considerably (compared to 1st gen).
- Much faster.
- Very much more reliable.

Disadvantages

- Still overheated quickly.
- Still had maintenance problems.

> **MEMORY HACK:** Transistor = "Tiny Replacement" for the vacuum tube. 2nd gen is the generation of FIRSTS: first symbolic language, first high-level languages (COBOL/FORTRAN), first stored-instruction memory.

> **TRAP:** Classic trick: "Which generation FIRST stored instructions in memory?" Students guess 1st gen because it sounds foundational. The real answer is 2ND gen. Lock it in.

3RD GENERATION (1964–1971) — Integrated Circuits

Core Facts

- The **INTEGRATED CIRCUIT (IC)** was the hallmark technology.
- Transistors were miniaturized and placed onto silicon chips, called **SEMICONDUCTORS**.
- Instead of punched cards and printouts, users interacted through **KEYBOARDS** and **MONITORS**, interfaced with an **OPERATING SYSTEM**.
- Allowed the device to run many applications at ONE time (multi-tasking begins here).

Advantages

- ICs are very small in size.
- Improved performance.
- Production cost cheaper.

Disadvantages

- ICs are sophisticated (complex) to manufacture/repair.

> **MEMORY HACK:** 3rd gen = "the KEYBOARD generation." Before this, humans fed computers punch cards; from here on, humans talk to computers directly via keyboard + monitor + OS.

> **TRAP:** Don't confuse INTEGRATED CIRCUIT (3rd) with MICROPROCESSOR (4th). An IC = transistors on ONE chip performing one function; a microprocessor puts the ENTIRE CPU on one chip. IC = 3rd gen. Microprocessor = 4th gen.

4TH GENERATION (1971–present) — Microprocessor

Core Facts

- The **MICROPROCESSOR** brought the fourth generation — thousands of integrated circuits built onto a SINGLE silicon chip.
- The **Intel 4004** chip (1971) located ALL components of the computer (CPU, memory, input/output controls) on one chip.
- Saw the development of **GUIs** (Graphical User Interfaces), the **MOUSE**, and **HANDHELD** devices.

> **MEMORY HACK:** 4th gen = "EVERYTHING ON ONE CHIP." Intel 4004 (1971) is the most exam-relevant date+name pair in this topic — memorize it as a pair, not separately.

> **TRAP:** Students treat "microprocessor" and "integrated circuit" as interchangeable. EVERY microprocessor contains ICs, but not every IC is a microprocessor. The exam tests this exact distinction.

5TH GENERATION (present and beyond) — Artificial Intelligence

Core Facts

- Based on **ARTIFICIAL INTELLIGENCE**.
- Still in development, though some applications exist — e.g. **VOICE RECOGNITION**.
- Uses **PARALLEL PROCESSING** and **SUPERCONDUCTORS** to make AI a reality.
- GOAL: devices that respond to **NATURAL LANGUAGE** input and are capable of **LEARNING** and **SELF-ORGANIZATION**.

MEMORY HACK: 5th gen = "the THINKING generation." Its goal (natural language + learning + self-organization) is what AI assistants aim to do today.

TRAP: Wrong answer alert: saying 5th gen is defined by "the internet" or "smartphones." It is defined by AI-based technology (parallel processing + superconductors + natural language), not any gadget.

Full Comparison Table

Gen	Years	Core Tech	Key Weakness
1st	1940–1956	Vacuum tubes, magnetic drums	Huge, hot, expensive, machine language only
2nd	1956–1963	Transistors	Still overheated; maintenance issues
3rd	1964–1971	Integrated Circuits (semiconductors)	ICs sophisticated to make/repair
4th	1971–present	Microprocessor (e.g. Intel 4004)	N/A — current dominant tech
5th	Present & beyond	Artificial Intelligence	Still largely in development

B COMPUTER NETWORKS

Chapter 3 — definition, classification, internetworks and hardware

3.1 What Is a Computer Network?

- A computer network is a group of interconnected computers. Connecting two or more computers together, primarily to share resources.
- Needs a medium of communication (physical connection) — wired or wireless.
- There must be at least **TWO** computer devices, called **WORKSTATIONS**. Each is a microcomputer (laptop, desktop, smartphone, palmtop etc.) or a supercomputer.
- Resources shared fall into two folds: **Software** (application and system software) and **Hardware** (printers, scanners, peripherals etc.).

> **MEMORY HACK:** 2 + Link + Share: at least 2 workstations, linked by a medium (wired/wireless), to share software and hardware resources.

3.2 Classification of Computer Networks

Networks are classified by four parameters:

Parameter	Meaning	Categories
Transmission medium	Hardware technology used to connect devices	Wired and Wireless
Functional relationship (architecture)	How elements relate to each other	Client-Server and Peer-to-Peer
Topology	Arrangement/mapping of links and nodes	Bus, Star, Ring
Geographical scalability	Area covered	PAN, LAN, CAN, MAN, WAN

> **MEMORY HACK:** M-F-T-G = Medium, Functional, Topology, Geographical. Four classification parameters.

3.2.1 By Transmission Medium

Type	Details
Wired	Uses physical wiring. Media: optical fibre and Ethernet (twisted pair and coaxial cables). Often deployed with hubs, switches, bridges and routers.
Wireless	Connects devices to hubs, switches and routers with no visible cables. Uses electromagnetic waves: radio, satellite, visible light, infrared, X-rays, gamma rays.

Electromagnetic radiation = energy propagated through space (and indirectly through solid objects) as advancing disturbance of electric and magnetic fields.

3.2.2 By Functional Relationship

Model	Key Points
Client-Server	Server is a centralized system connecting several clients. The more simultaneous clients, the more resources the server needs.
Peer-to-Peer (P2P)	Two or more computers (peers) pool resources and communicate in a decentralized system. Peers are co-equal nodes in a non-hierarchical network.

> **MEMORY HACK:** Client-Server = Central boss. Peer-to-Peer = Pals with equal power.

3.2.3 By Topology

Network topology = study of the arrangement or mapping of the elements (links and nodes), especially the physical (real) and logical (virtual) interconnections between nodes. Basic types: **Bus, Star, Ring**.

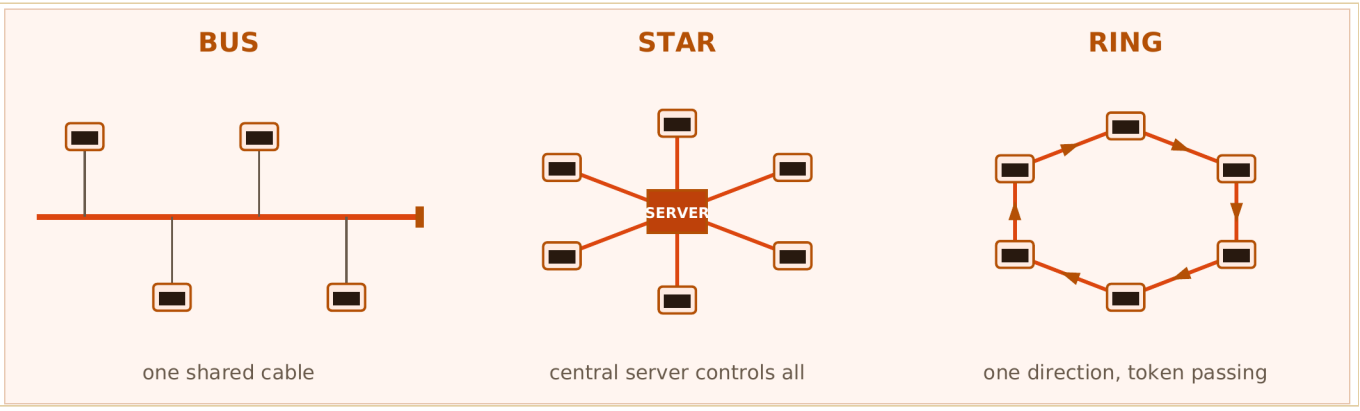


Fig. B1: Bus, Star and Ring topologies

Topology	How it works	Exam facts
Bus	Local computers share one single bus (communication channel) — a linear coaxial cable into which devices tap.	Connects through a **tap** (a **passive** device: does not alter the signal, needs no electricity). Workstation end has a **NIC**. When one device transmits, ALL others receive.
Star	End-user computers tied to a central computer (typically the server) which controls transmissions of all other workstations.	Everything passes through the centre.
Ring	Local computers tied in a closed loop and transmit on an equal basis using **tokens**.	Supports only ONE channel of information, flowing in ONE direction. Data that has circled must be removed, otherwise it keeps circling. The **original transmitting workstation** removes it.

MEMORY HACK: BUS = one Backbone cable. STAR = Server at the Centre. RING = Rotating token, one way.

TRAP: A bus tap is PASSIVE (no electricity, signal unchanged). The NIC is what lets the workstation send and receive. In a ring, the sender removes the circling data, not the receiver.

3.2.4 By Geographical Scalability

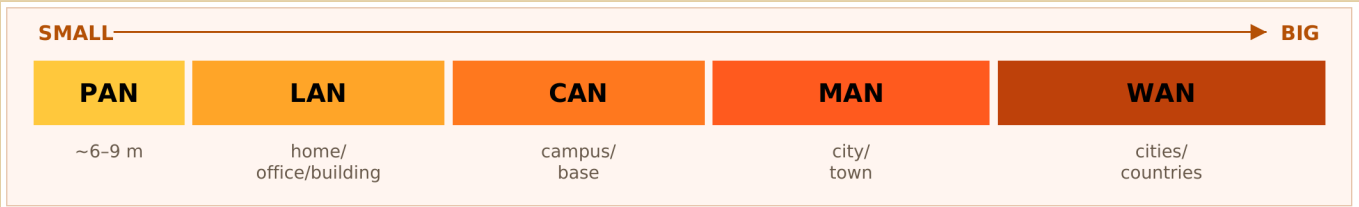


Fig. B2: Network types from smallest to largest

Type	Definition and key facts
PAN Personal Area Network	Communication among devices **close to one person**. Devices: printers, fax machine, telephones, PDAs, scanners. Reach about **20–30 feet (6–9 m)**. Wired with buses like **USB** and **FireWire**. Wireless version = **WPAN** via **Infrared** and **Bluetooth**.
LAN Local Area Network	Interconnects a variety of data communication devices in a **small area** (home, office, building) and broadcasts data at **high transfer rates**. Typically based on **Ethernet**. Wired: twisted pair CAT 5/CAT 6; servers use CAT 5e/6e supporting IEEE 802.3 at 1 Gb/s. Wireless: **802.11b / 802.11g**. Most LANs run **10–100 Mb/s**; 802.3 goes up to **10 Gb/s**. Versus WANs: higher speed, smaller range, **no leased telecom lines**.
CAN Campus Area Network	Connects **two or more LANs** within a specific, contiguous area (college campus, industrial complex, military base). Can be seen as a type of MAN but smaller than a typical MAN.
MAN Metropolitan Area Network	Connects two or more LANs or CANs but does **not extend beyond the metropolis/town/city**. Built by connecting routers, switches and hubs.
WAN Wide Area Network	Covers a **broad geographic area** (city to city, country to country). Often uses transmission facilities from common carriers such as telephone companies.

MEMORY HACK: PAN, LAN, CAN, MAN, WAN = "Pretty Little Cats Meow Wildly" — smallest to biggest. LAN is the one with Ethernet + highest speeds.

TRAP: A CAN connects LANs in ONE contiguous place (campus). A MAN stays inside one city. WAN crosses cities/countries. Also: LANs do NOT need leased telecom lines; WANs often do.

3.3 Internetwork

- Two or more networks (or segments) connected using a device such as a **router** = an internetwork.
- Interconnection of public, private, commercial or governmental networks also counts. Modern practice uses the **Internet Protocol**.
- Three variants (by who administers and who participates): **Intranet, Extranet, Internet**.
- If an intranet/extranet connects to the internet, it is normally protected from unauthorised access. The internet itself is NOT part of the intranet or extranet.

	Intranet	Extranet	Internet
What is it	Set of interconnected networks using IP and IP tools (web browsers, FTP) under ONE administrative entity.	Network limited to a single organisation but with limited connections to networks of other (usually trusted) organisations.	Collection of ALL networks (LAN, MAN, WAN etc.), thousands of networks and attached devices.
Access	Closed to the rest of the world; only specific users. Usually a company's internal network.	Clients may be given access to part of an intranet. Customers may not be fully "trusted" for security.	Open, public.
Key fact	Large intranets have their own web server.	By definition it cannot be a single LAN; must have at least one connection to an external network. Can be a CAN, MAN, WAN etc.	Began as ARPANET (U.S. Department of Defence, 1960s).

MEMORY HACK: INtranet = INside only. EXtranet = EXtra access for outsiders (clients/partners). INTERnet = everyone, INTERconnected.

TRAP: An extranet can NEVER be just one LAN. And the internet is not part of an intranet or extranet, though it can serve as a portal to parts of an extranet.

3.3.3 The Internet in Detail

- Began as **ARPANET** (Advanced Research Projects Agency NETwork), developed by the U.S. Department of Defence in the 1960s.
- One high-speed backbone linked university, government and research sites; supported **56 kbps**; later financed by the National Science Foundation (**NSF**).
- A **protocol** = a set of rules determining the format and transmission of data.
- Protocols: **IP** (Internet Protocol), **TCP** (Transmission Control Protocol), **ARP** (Address Resolution Protocol), **DHCP** (Dynamic Host Configuration Protocol), **NAT** (Network Address Translation).
- Service providers and large enterprises exchange reachability of address ranges through **BGP** (Border Gateway Protocol).

World Wide Web (WWW)

- A vast collection of electronic documents on many Web servers, containing text and images, accessed by clicking links in a browser.
- Web pages are created with **HTML** (HyperText Markup Language) — by hand in a text editor like Notepad, or with a page-authoring tool.
- Store the page on a computer with Web server software and an internet connection. The server accepts **HTTP** requests from browsers, retrieves the page and returns it.

> **MEMORY HACK:** Internet = the road network. WWW = the shops on the road (web pages). HTML builds the page, HTTP delivers it. ARPANET → 1960s → 56 kbps.

> **TRAP:** The Internet and the WWW are NOT the same thing. The Internet is the collection of networks; the WWW is one service (the collection of documents) that runs on it.

3.4 Basic Hardware Components Used in Networking

Device	What it does	Exam facts
NIC Network Interface Card	Hardware that lets a computer communicate over a network. Provides physical access to a networking medium.	Gives low-level addressing through **MAC** (Media Access Control) addresses. Works via cable or wireless.
Bridge	Connects multiple network segments to create an aggregate network.	Distinct from routing (which keeps networks communicating as separate networks). **4 types**: simple, multiport, learning/transparent, source route.
Hub	Has multiple ports. A packet arriving at one port is copied to **ALL** ports.	Rudimentary: simply copies data to all connected nodes. Destination address in the frame does NOT change to a broadcast address.
Switch	Forwards and filters data using **MAC addresses** in the packet.	Unlike a hub, forwards only to the ports involved in the communication.
Router	Forwards data packets **between networks** using headers and forwarding tables to find the best path.	Connected to at least two networks (two LANs/WANs, or a LAN and its ISP's network).

MEMORY HACK: Hub = Shouts to everyone. Switch = Whispers only to the right port (by MAC). Router = Road map between different networks. NIC = the computer's own "door" with a MAC name tag.

TRAP: Hub vs Switch is the favourite: hub copies to ALL ports, switch sends only to the ports involved. Bridge connects SEGMENTS (not routing); router connects NETWORKS. A port is the circuit (hardware + circuitry) that links one device with another.

C

NETWORK SECURITY & MALWARE

Chapter 3 — protecting information and infrastructure

3.5 Computer Network Security

- Security = the state of well-being of information and infrastructure in which successful, undetected theft, tampering and disruption of information and services is kept low or tolerable.
- Simply put: the protection of computing systems and the data they store or access.
- Objective: protect information and property from theft, corruption or natural disaster, while keeping it accessible and productive to intended users.
- Small home/office needs basic security; large businesses need high maintenance and advanced software/hardware against hackers/spammers.

Authentication Factors

Level	What is used	Examples
One-factor	Something you **KNOW**	Username + password
Two-factor	Something you **HAVE** is added	Security token or "dongle", ATM card, mobile phone
Three-factor	Something **ON YOUR BODY**	Fingerprint, retinal scan

MEMORY HACK: 1 = KNOW (password). 2 = + HAVE (token/card/phone). 3 = + ARE (fingerprint/retina). "Know, Have, Body."

Security Concepts: CIA + AAN

Group	Concept	Meaning / loss
Information	Confidentiality	Only authorised people read/copy information. Loss of confidentiality = read or copied by someone not authorised. Crucial for research data, medical and insurance records, new product specs, investment strategies.
Information	Integrity	Information is not changed in unexpected ways. Loss of integrity = unauthorised changes (human error or deliberate tampering). Crucial for electronic funds transfers, air traffic control, financial accounting.
Information	Availability	Information stays accessible to intended users.
People	Authentication	Proving who the user is.
People	Authorization	What that user is allowed to do.
People	Non-repudiation	User cannot deny an action they performed.

MEMORY HACK: CIA protects the information; AAN concerns the people who use it. "CIA for data, AAN for people."

TRAP: Confidentiality failure = someone READ/COPIED it. Integrity failure = someone CHANGED it. Legal privacy duties apply to banks, loan companies, hospitals, doctors, medical labs, counselling and drug-treatment agencies, and tax agencies.

3.6 Computer Viruses and Other Destructive Programs

Malware is the general name for ALL programs harmful to a computer system: viruses, Trojans, worms and spyware (plus logic bombs in the notes). Viruses and malware are names for all forms of destructive programs.

Type	Definition	Key facts
Virus	Program that attaches to, overwrites or replaces another program to reproduce itself without the user's knowledge.	Must be executed to operate. Classified by how they infect and what they infect. Often harmless for years; may randomly damage files, and over time destroy files and disks.
Trojan Horse	Destructive program disguised inside an innocent-looking program (games, attractive software the victim is tempted to run).	**NOT a virus**: it does NOT reproduce. Worms/viruses may hide inside it. Can send information to its developer, who may control the computer. Used for embezzlement and industrial espionage; can self-destruct. Named after the Greek wooden horse at Troy (Odysseus).
Logic Bomb	Like a Trojan but with a **timing device**: goes off at a particular date/time (or trigger).	Example: **Michelangelo** virus. Does not need to reproduce. Favoured by disgruntled ex-employees (trigger: name deleted from payroll) and by suppliers/consultants as "insurance" if bills are unpaid.
Spyware	Programs that attach to the OS in nefarious ways, tracking internet habits, showing unwanted sales offers and generating traffic for host sites.	Usually not designed to damage the computer. Enters without permission, hides in the background, serves targeted ads, redirects browsers. Estimated **80%+** of PCs infected. Drains processing power. Not a virus because it does not replicate.
Worm	Program that spreads usually over **network connections**.	Mostly found in multi-user systems, not normally on one-person computers.

MEMORY HACK: Virus = Very good at copying. Trojan = Trick (disguise). Logic bomb = Later (timer). Spyware = Snooper (ads and tracking). Worm = Wires (spreads over the network).

TRAP: Key test for any suspect program: can it reproduce? Only VIRUSES copy themselves and attach the copy to new files. Trojans, logic bombs and spyware do not. Also: spyware is NOT a virus.

TRAP: Heads up on the worm line: the lecturer's note says worms "always need a host program to spread" while contrasting with a virus. Standard textbooks say the opposite: a virus needs a host program, a worm does NOT (it spreads on its own over networks). If a question uses this contrast, answer with the standard definition and be ready to mention both.

3.7 Countermeasures for Preventing Network Threats

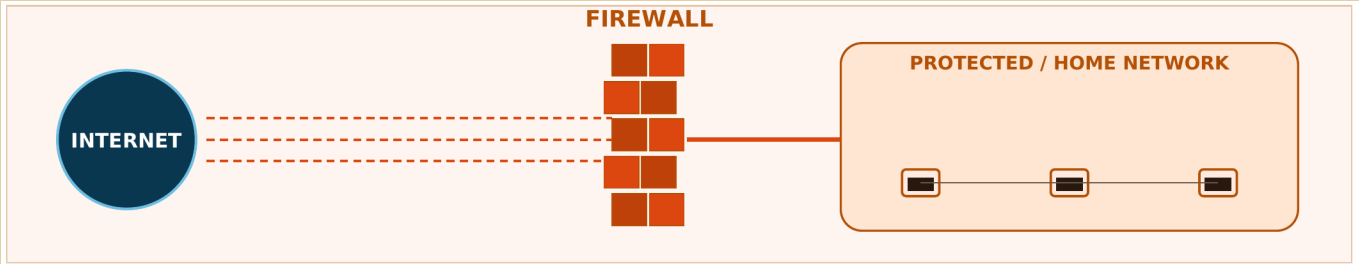


Fig. C1: A firewall sits between the Internet and the protected network

Measure	Facts
Firewall	A component or set of components that restricts access between a protected network and the Internet (or between other sets of networks). Protects against offensive websites and potential hackers. Works like a physical firewall that stops fire spreading from one area to the next. Comes in **hardware** and **software** varieties.
Anti-virus	Must be installed, running actively and **updated**. New viruses are found almost daily. Software not updated for a month or more is equivalent to no antivirus. Vendors analyse new threats (file size, specific text, message body/subject line, how the file works) and create a **signature** (footprint) shipped in update files. Most vendors update virus definitions at least weekly.

Antivirus Products Named in the Notes

Paid / commercial	Free options
McAfee ViruScan (Network Associates), Norton Antivirus (Symantec), Sophos, F-Secure	Avira, Antivir, AVG

MEMORY HACK: Firewall = the wall that keeps destructive forces out (hardware or software). Antivirus = the vaccine: it only knows threats whose signatures it has, so keep it UPDATED.

TRAP: Firewall blocks unauthorised ACCESS between networks. Antivirus detects KNOWN malicious programs through signatures. Outdated antivirus (a month+) = useless.

D PSEUDOCODE

Plan the logic in human language before you code

- Pseudo-code is an informal, high-level description of the operating principle of an algorithm.
- Uses the structural conventions of a programming language but is meant for human reading, not machine reading.
- Typically omits non-essential details: variable declarations, system-specific code and some subroutines.
- Programming language is augmented with natural-language description or compact mathematical notation.
- Purpose: easier for people to understand than real code; an efficient, environment-independent description of an algorithm's key principles.
- Commonly used in textbooks and scientific publications, and in planning program development (sketching structure before coding).

Key Characteristics

Point	Detail
Standard	No standard syntax exists. It is NOT an executable program.
Not the same as	Skeleton programs / dummy code, which CAN be compiled without errors.
Graphical alternatives	Flowcharts and UML (Unified Modelling Language) charts — but more spacious on paper.
Syntax sources	Writers borrow control structures from Pascal, BASIC, C, C++, Java, Lisp and ALGOL.
Style range	From a near-exact imitation of a programming language to formatted prose.
Common habit	Variable declarations omitted; function calls and code blocks (e.g. inside a loop) replaced by a one-line natural-language sentence.

MEMORY HACK: Pseudo-code = "Fake code for Humans". Pseudo means fake: it will NOT run, no strict rules, written for people. Remember PBALC as syntax sources: Pascal, BASIC, ALGOL, Lisp, C-family.

TRAP: Pseudo-code is NOT a skeleton program. Skeleton/dummy code compiles; pseudo-code never does.

Worked Example: Average of N Numbers (ANSI-BASIC style)

```
10   Comment: Pseudo-code to find average of N numbers
20   Initialise counter = 0
30   Initialise sum = 0
40   INPUT N
50   INPUT number
60   sum = sum + number
70   counter = counter + 1
80   IF (counter < N) THEN GOTO line 50
90   average = sum / N
100  PRINT average
110  END
```

How it works

- Lines 20–30: set counter and sum to 0 (starting values).
- Line 40: read N, how many numbers there are.
- Lines 50–70: read a number, add it to sum, add 1 to counter.
- Line 80: loop back to line 50 while the counter is still less than N.
- Lines 90–100: average = sum ÷ N, then print it. Line 110: END.

MEMORY HACK: Average = SUM ÷ N. Loop skeleton: Start at 0 → Input → Add to sum → Count up → Repeat until N → Divide → Print.

TRAP: The original handout writes line 80 as IF (i < N), but the program only has a variable called counter (no "i"). It is a typo in the source: the loop test should compare counter with N. Write it with counter in the exam.

E OSI MODEL

Seven layers that split network communication into jobs

> *This part is standard textbook content added for revision. It is not in the lecturer's handout.*

What Is the OSI Model?

- **OSI** = **Open Systems Interconnection**. It is a reference model published by the **ISO** (International Organization for Standardization).
- It splits network communication into **SEVEN layers**. Each layer has one job and talks to the layer above and below it.
- It is a guide for understanding networks. It is not a device, a cable or a product.
- Data going out starts at layer 7 and moves DOWN to layer 1. Data coming in moves UP from layer 1 to layer 7.

The Seven Layers — Top to Bottom

7	Application	Data
6	Presentation	Data
5	Session	Data
4	Transport	Segment
3	Network	Packet
2	Data Link	Frame
1	Physical	Bit

No.	Layer	Its job	Data unit	Examples
7	**Application**	Gives network services to the user's programs: web, email, file transfer.	Data	HTTP, FTP, SMTP, DNS
6	**Presentation**	Translates data format; handles encryption and compression.	Data	Encryption, compression
5	**Session**	Opens, manages and closes a session between two applications.	Data	Session set-up and tear-down
4	**Transport**	End-to-end delivery between two hosts; breaks data into segments.	Segment	TCP, UDP
3	**Network**	Logical (IP) addressing and routing between networks; finds the best path.	Packet	IP, router
2	**Data Link**	Delivers frames on the same network using MAC addresses; error detection.	Frame	MAC address, NIC, switch, bridge
1	**Physical**	Sends raw bits as signals over the cable or air.	Bit	Cables, hub, radio signals

MEMORY HACK: **Bottom to top (1 → 7):** "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way" = Physical, Data Link, Network, Transport, Session, Presentation, Application.
Top to bottom (7 → 1): "**A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing" = Application, Presentation, Session, Transport, Network, Data Link, Physical.

TRAP: Students swap **Session (5)** and **Transport (4)**. Transport is the LOWER one. Another favourite: putting encryption at the Application layer. Encryption and data format belong to **Presentation (6)**.

Layers 1 to 3 — The Lower Layers

Layer	What to know
1 · Physical	Deals with **bits**, cables and signals. Nothing is read or understood here, it is only sent. A **hub** works here because it only copies the signal to all ports.
2 · Data Link	Uses **MAC addresses** and sends **frames** between devices on the same network. The **NIC**, **switch** and **bridge** work here. Detects errors in frames.
3 · Network	Uses **IP addresses** and sends **packets** between different networks. **Routing** (finding the best path) happens here. The **router** works here.

Layers 4 to 7 — The Upper Layers

Layer	What to know
4 · Transport	Makes sure data gets from one host to the other end to end. Breaks data into **segments**. **TCP** (Transmission Control Protocol) works here.
5 · Session	**Opens, manages and closes** a session between two applications, like starting, keeping and ending a phone call.
6 · Presentation	Translates data so both sides understand it. Handles **encryption** and **compression**.
7 · Application	The layer closest to the user. Web browsing, email and file transfer services. **HTTP** works here.

Link to Your Hardware and Protocols (Part B)

Item from Part B	OSI layer	Why
Hub	1 · Physical	Only copies the signal to all ports.
NIC / MAC address	2 · Data Link	MAC addresses belong to layer 2.
Switch / Bridge	2 · Data Link	Forward using MAC addresses.
Router	3 · Network	Forwards packets between networks using IP and finds the best path.
IP (Internet Protocol)	3 · Network	Gives logical addresses and routes packets.
TCP	4 · Transport	Controls end-to-end delivery.

Item from Part B	OSI layer	Why
HTTP	7 · Application	Carries web page requests from browser to web server.

MEMORY HACK: **Hub 1, Switch 2, Router 3.** The three network boxes sit on the first three layers in order of how clever they are. **IP = 3, TCP = 4, HTTP = 7.** Count the data units from the bottom: **Bit, Frame, Packet, Segment, then Data.** (B-F-P-S-D)

TRAP: Do not say a switch works at the Network layer. A **switch is layer 2** (MAC) and a **router is layer 3** (IP). Also, the TCP/IP model groups these seven layers into fewer layers (commonly four), so a question may ask for one or the other. Read which model it names.

F MASTER RAPID-FIRE SHEET

Likely exam questions with quick answers

F1. Generations of Computers

Question	Answer
What technology defined the 1st generation?	Vacuum tubes (circuitry) and magnetic drums (memory)
Name two examples of first-generation computers.	UNIVAC and ENIAC
What replaced vacuum tubes in the 2nd generation?	Transistors
Which generation first stored instructions in memory?	2nd generation
Two early high-level languages of the 2nd generation?	COBOL and FORTRAN
Hallmark technology of the 3rd generation?	The Integrated Circuit (IC)
What are transistors on a silicon chip called?	Semiconductors
How did users interact with 3rd-gen computers?	Keyboards and monitors, interfaced with an operating system
What brought about the 4th generation?	The microprocessor
Which 1971 chip put all components on one chip?	The Intel 4004
Two input innovations of the 4th generation?	GUIs and the mouse

Question	Answer
What is the 5th generation based on?	Artificial Intelligence (AI)
Two technologies enabling 5th-gen AI?	Parallel processing and superconductors
Ultimate goal of 5th-generation computing?	Devices that respond to natural language and are capable of learning and self-organization

F2. Computer Networks

Question	Answer
Minimum number of workstations for a network?	Two
Two types of resources shared on a network?	Software and hardware
Four parameters for classifying networks?	Transmission medium, functional relationship, topology, geographical scalability
Two functional (architecture) types?	Client-Server and Peer-to-Peer
Three basic topologies?	Bus, Star, Ring
Is a bus tap active or passive?	Passive (no electricity, does not alter the signal)
Which topology uses tokens and one-direction flow?	Ring
Which device removes circling data in a ring?	The workstation that originally transmitted it
Reach of a PAN?	About 20–30 feet (6–9 m)

Question	Answer
Wireless PAN technologies?	Infrared and Bluetooth (WPAN)
Technology LANs are typically based on?	Ethernet
Network connecting LANs within a campus?	CAN (Campus Area Network)
Network that stays within a city or town?	MAN (Metropolitan Area Network)
Network covering cities or countries?	WAN (Wide Area Network)
What connects two or more networks to form an internetwork?	A router
Three variants of internetwork?	Intranet, Extranet, Internet
Can an extranet be a single LAN?	No — it needs at least one connection to an external network
What was the Internet's forerunner?	ARPANET (U.S. Department of Defence, 1960s)
Which language builds web pages?	HTML (HyperText Markup Language)
Which protocol carries web page requests to a web server?	HTTP
Device that copies data to ALL ports?	Hub
Device that forwards using MAC addresses to specific ports only?	Switch

Question	Answer
Device that forwards packets between networks using forwarding tables?	Router
What gives a computer a MAC address and network access?	The Network Interface Card (NIC)

F3. Security, Malware and Pseudocode

Question	Answer
Three basic security concepts for information?	Confidentiality, Integrity, Availability
Three security concepts for the people using information?	Authentication, Authorization, Non-repudiation
Loss of confidentiality means?	Information read or copied by someone not authorised
Loss of integrity means?	Information modified in unexpected (unauthorised) ways
What does two-factor authentication add?	Something you have (token, ATM card, phone)
What does three-factor authentication use?	Something on your body (fingerprint, retinal scan)
General name for all harmful programs?	Malware
Which malware reproduces itself?	The virus (only viruses copy themselves)
Is a Trojan horse a virus?	No — it does not reproduce
Which malware includes a timing device?	Logic bomb (e.g. Michelangelo)

Question	Answer
Which malware tracks internet habits and shows targeted ads?	Spyware
Which malware spreads over network connections?	Worm
What restricts access between a protected network and the Internet?	Firewall (hardware or software)
How long without updates makes antivirus useless?	A month or more
What identifies a threat in antivirus update files?	A signature (footprint)
Name two free antivirus products from the notes.	Avira and AVG (also Antivir)
What is pseudo-code?	Informal high-level description of an algorithm for humans, not machines
Is pseudo-code executable?	No
Graphical alternatives to pseudo-code?	Flowcharts and UML charts
Formula used in the average example?	average = sum / N

F4. OSI Model

Question	Answer
What does OSI stand for?	Open Systems Interconnection
How many layers does the OSI model have?	Seven

Question	Answer
Which body published the OSI model?	ISO
Which layer is at the bottom? At the top?	Physical (1) at the bottom, Application (7) at the top
Layers in order from 1 to 7?	Physical, Data Link, Network, Transport, Session, Presentation, Application
Memory line from layer 1 up?	Please Do Not Throw Sausage Pizza Away
Memory line from layer 7 down?	All People Seem To Need Data Processing
Which layer uses MAC addresses?	Data Link (layer 2)
Which layer does routing and IP addressing?	Network (layer 3)
Which layer gives end-to-end delivery?	Transport (layer 4)
Which layer handles encryption and data format?	Presentation (layer 6)
Which layer opens, manages and closes sessions?	Session (layer 5)
Data units from layer 1 up to layer 4?	Bit, Frame, Packet, Segment
At which layers do a hub, switch and router work?	Hub: Physical (1). Switch: Data Link (2). Router: Network (3)
Which layers do IP, TCP and HTTP belong to?	IP: Network (3). TCP: Transport (4). HTTP: Application (7)